Class: **CAPE Pure Mathematics Unit 1**

Topic: **Past Paper June 2015 – Paper 3 – Solutions**

June 2015 – Question 1 – Solution

(a) Let $\boldsymbol{p}$ and $\boldsymbol{q}$ be two propositions.

(i) Copy and complete the truth table below. [5]

$\boldsymbol{p}$	$\boldsymbol{q}$	$\sim\boldsymbol{p}$	$\sim\boldsymbol{q}$	$\boldsymbol{p} \vee \boldsymbol{q}$	$\sim\boldsymbol{p} \wedge \sim\boldsymbol{q}$	$(\boldsymbol{p} \vee \boldsymbol{q}) \vee (\sim\boldsymbol{p} \wedge \sim\boldsymbol{q})$
T	T	F	F	T	F	T
T	F	F	T	T	F	T
F	T	T	F	T	F	T
F	F	T	T	F	T	T

(ii) Hence, or otherwise, state if $(\boldsymbol{p} \vee \boldsymbol{q}) \vee (\sim\boldsymbol{p} \wedge \sim\boldsymbol{q})$ is a tautology.

Justify your response. [1]

The statement $(\boldsymbol{p} \vee \boldsymbol{q}) \vee (\sim\boldsymbol{p} \wedge \sim\boldsymbol{q})$ is a tautology since all of the truth values are true.

(b) Solve the equation $\log_4(x^2+1) - \log_2(2x-1) = 0$. [8]

$$\log_4(x^2+1) - \log_2(2x-1) = 0$$

$$\frac{\log_2(x^2+1)}{\log_2 4} - \log_2(2x-1) = 0$$

$$\frac{\log_2(x^2+1)}{2} - \log_2(2x-1) = 0$$

$$\log_2(x^2+1) - 2\log_2(2x-1) = 0$$

$$\log_2(x^2+1) - \log_2(2x-1)^2 = 0$$

$$\log_2 \frac{(x^2+1)}{(2x-1)^2} = 0$$

Converting to exponential form gives:

$$\frac{(x^2+1)}{(2x-1)^2} = 2^0$$

$$\frac{(x^2+1)}{(2x-1)^2} = 1$$

$$(x^2+1) = (2x-1)^2$$

$$x^2+1 = 4x^2-4x+1$$

$$0 = 4x^2 - x^2 - 4x + 1 - 1$$

$$0 = 3x^2 - 4x$$

$$0 = x(3x-4)$$

Either $x = 0$ (invalid) or $3x - 4 = 0$

$$3x = 4$$

$$x = \frac{4}{3}$$

$$\therefore x = \frac{4}{3}$$

(c) Let $x, y, k \in \mathbb{R}$. An operation $*$ is defined as $x * y = \frac{xy}{x+y-k}$ where $x + y \neq k$.

Show that $*$ is associative. [6]

Let $x, y, z \in \mathbb{R}$.

For $*$ to be associative, $(x * y) * z = x * (y * z)$.

Taking L.H.S:

$$(x * y) * z = \left(\frac{xy}{x+y-k}\right) * z$$

$$= \frac{\left(\frac{xy}{x+y-k}\right)z}{\left(\frac{xy}{x+y-k}\right)+z-k}$$

$$= \frac{\left(\frac{xyz}{x+y-k}\right)}{\left(\frac{xy}{x+y-k}+z-k\right)}$$

$$= \left(\frac{xyz}{x+y-k}\right) \div \left(\frac{xy}{x+y-k}+z-k\right)$$

$$= \left(\frac{xyz}{x+y-k}\right) \div \left[\frac{xy+z(x+y-k)-k(x+y-k)}{x+y-k}\right]$$

$$= \left(\frac{xyz}{x+y-k}\right) \div \left(\frac{xy+xz+yz-kz-kx-ky+k^2}{x+y-k}\right)$$

$$= \frac{xyz}{x+y-k} \times \frac{x+y-k}{xy+xz+yz-kz-kx-ky+k^2}$$

$$= \frac{xyz}{xy+xz+yz-kz-kx-ky+k^2} \qquad \rightarrow \text{Expression 1}$$

Taking R.H.S:

$$x * (y * z) = x * \left(\frac{yz}{y+z-k}\right)$$

$$= \frac{x\left(\frac{yz}{y+z-k}\right)}{x+\left(\frac{yz}{y+z-k}\right)-k}$$

$$= \frac{\left(\frac{xyz}{y+z-k}\right)}{\left(x+\frac{yz}{y+z-k}-k\right)}$$

$$= \left(\frac{xyz}{y+z-k}\right) \div \left(x + \frac{yz}{y+z-k} - k\right)$$

$$= \left(\frac{xyz}{y+z-k}\right) \div \left[\frac{x(y+z-k)+yz-k(y+z-k)}{y+z-k}\right]$$

$$= \left(\frac{xyz}{y+z-k}\right) \div \left(\frac{xy+xz-kx+yz-ky-kz+k^2}{y+z-k}\right)$$

$$= \frac{xyz}{y+z-k} \times \frac{y+z-k}{xy+xz-kx+yz-ky-kz+k^2}$$

$$= \frac{xyz}{xy+xz-kx+yz-ky-kz+k^2}$$

$$= \frac{xyz}{xy+xz+yz-kz-kx-ky+k^2} \qquad \rightarrow \text{Expression 2}$$

Since Expression 1 and Expression 2, then $(x * y) * z = x * (y * z)$.

$\therefore$ The operation $*$ is associative.

Q.E.D.

Total: 20 marks

June 2015 – Question 2 – Solution

(a) $ABCD$ is a parallelogram such that A has coordinates $(0, 0)$. The position vectors B and D are $\boldsymbol{a}$ and $\boldsymbol{b}$ respectively. Using vector algebra, show that

$$(AC)^2 + (BD)^2 = (AB)^2 + (BC)^2 + (DC)^2 + (AD)^2 \qquad [7]$$

A has coordinates $(0, 0)$.

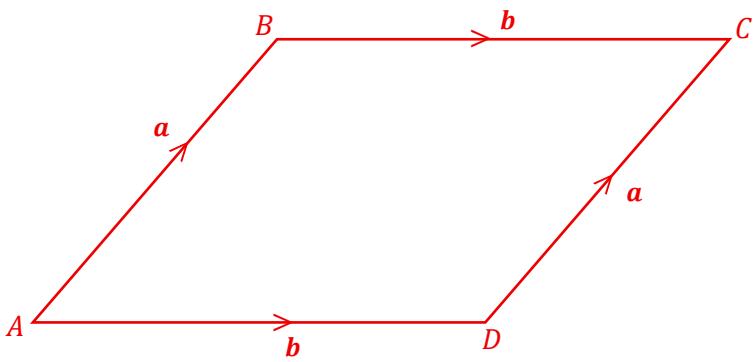
$\overrightarrow{OA} = \mathbf{0}$

The position vectors B and D are $\boldsymbol{a}$ and $\boldsymbol{b}$ respectively.

$\overrightarrow{OB} = \boldsymbol{a}$ and $\overrightarrow{OD} = \boldsymbol{b}$

Since $ABCD$ is a paralellogram, then $\overrightarrow{DC} = \boldsymbol{a}$ and $\overrightarrow{BC} = \boldsymbol{b}$.

Consider the diagram below.



Using triangle law,

$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA}$ and $\overrightarrow{AD} = \overrightarrow{OD} - \overrightarrow{OA}$

$= \boldsymbol{a} - \mathbf{0}$ $\qquad$ $= \boldsymbol{b} - \mathbf{0}$

$= \boldsymbol{a}$ $\qquad$ $= \boldsymbol{b}$

Using triangle law,

$\overrightarrow{BD} = \overrightarrow{OD} - \overrightarrow{OB}$ and $\overrightarrow{AC} = \overrightarrow{AB} + \overrightarrow{BC}$

$= \boldsymbol{b} - \boldsymbol{a}$ $= \boldsymbol{a} + \boldsymbol{b}$

Now,

Taking L.H.S:

$$(AC)^2 + (BD)^2 = (\boldsymbol{a} + \boldsymbol{b})^2 + (\boldsymbol{b} - \boldsymbol{a})^2$$

$$= \boldsymbol{a}^2 + 2\boldsymbol{ab} + \boldsymbol{b}^2 + \boldsymbol{b}^2 - 2\boldsymbol{ab} + \boldsymbol{a}^2$$

$$= 2\boldsymbol{a}^2 + 2\boldsymbol{b}^2$$

Taking R.H.S:

$$(AB)^2 + (BC)^2 + (DC)^2 + (AD)^2 = \boldsymbol{a}^2 + \boldsymbol{b}^2 + \boldsymbol{a}^2 + \boldsymbol{b}^2$$

$$= 2\boldsymbol{a}^2 + 2\boldsymbol{b}^2$$

$$= \text{L.H.S.}$$

$$\therefore (AC)^2 + (BD)^2 = (AB)^2 + (BC)^2 + (DC)^2 + (AD)^2$$

Q.E.D.

(b) The equation of a circle, C_1, is given by $x^2 + y^2 - 4x - 6y + 3 = 0$.

Find the equation of the tangent to C_1 at the point $(5, 2)$. [6]

$$x^2 + y^2 - 4x - 6y + 3 = 0$$

$$(x^2 - 4x) + (y^2 - 6y) = -3$$

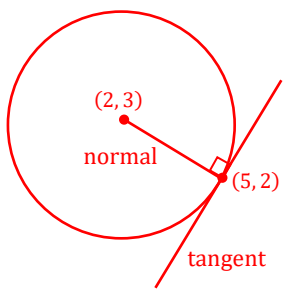
$$(x^2 - 4x + 4) + (y^2 - 6y + 9) = -3 + 4 + 9$$

$$(x - 2)^2 + (y - 3)^2 = 10$$

which is in the form $(x - a)^2 + (y - b)^2 = r^2$,

where centre is $C(2, 3)$ and radius, $r = \sqrt{10}$ units.

Consider the diagram below:



Points are $(2, 3)$ and $(5, 2)$.

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

$$= \frac{2-3}{5-2}$$

$$= \frac{-1}{3}$$

Gradient of normal $= -\frac{1}{3}$

Gradient of tangent $= 3$

Substituting $m = 3$ and point $(5, 2)$ into $y - y_1 = m(x - x_1)$ gives:

$y - 2 = 3(x - 5)$

$y - 2 = 3x - 15$

$y = 3x - 15 + 2$

$y = 3x - 13$

$\therefore$ The equation of the tangent to C_1 at the point $(5, 2)$ is $y = 3x - 13$.

(c) Angles $2A$ and $2B$ are obtuse angles with $\sin(2A) = \frac{3}{5}$ and $\cos(B) = \frac{4}{5}$.

Without calculating the value of A and B, find the EXACT value of

(i) $\cos(2A)$ [2]

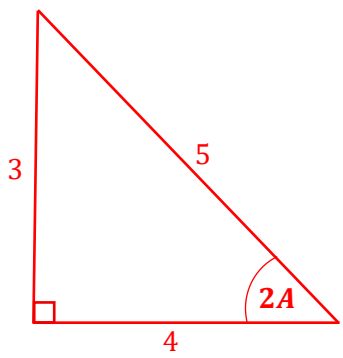
Since Angle $2A$ is obtuse, then we are looking at the second quadrant.

We are given that:

$$\sin(2A) = \frac{3}{5}$$

$$= \frac{opp}{hyp}$$

Consider the diagram below.



Cosine is negative in the second quadrant.

Now,

$$\cos(2A) = \frac{adj}{hyp}$$

$$= \frac{-4}{5}$$

$\therefore$ The value of $\cos(2A) = -\frac{4}{5}$.

(ii) $\cos(A+B)$ [5]

We are given that $\cos(B) = \frac{4}{5}$.

So, we have,

$\sin^2 B = 1 - \cos^2 B$

$\sin^2 B = 1 - \left(\frac{4}{5}\right)^2$

$\sin^2 B = 1 - \frac{16}{25}$

$\sin^2 B = \frac{9}{25}$

$\sin B = \sqrt{\frac{9}{25}}$

$\sin B = \frac{3}{5}$

Also,

$\cos(2A) = -\frac{4}{5}$

$1 - 2\cos^2 A = -\frac{4}{5}$

$-2\cos^2 A = -\frac{4}{5} - 1$

$-2\cos^2 A = -\frac{9}{5}$

$\cos^2 A = -\frac{9}{5} \times -\frac{1}{2}$

$\cos^2 A = \frac{9}{10}$

$\cos A = \sqrt{\frac{9}{10}}$

$\cos A = \frac{3}{\sqrt{10}}$

We have:

$$\sin(2A) = \frac{3}{5}$$

$$2\sin A\cos A = \frac{3}{5}$$

$$2\sin A\left(\frac{3}{\sqrt{10}}\right) = \frac{3}{5}$$

$$\left(\frac{6}{\sqrt{10}}\right)\sin A = \frac{3}{5}$$

$$\sin A = \frac{3}{5} \div \frac{6}{\sqrt{10}}$$

$$\sin A = \frac{3}{5} \times \frac{\sqrt{10}}{6}$$

$$\sin A = \frac{\sqrt{10}}{10}$$

$$\sin A = \frac{\sqrt{10}}{\sqrt{10}\sqrt{10}}$$

$$\sin A = \frac{1}{\sqrt{10}}$$

Using the addition formula,

$$\cos(A+B) = \cos A\cos B - \sin A\sin B$$

$$= \left(\frac{3}{\sqrt{10}}\right)\left(\frac{4}{5}\right) - \left(\frac{1}{\sqrt{10}}\right)\left(\frac{3}{5}\right)$$

$$= \frac{12}{5\sqrt{10}} - \frac{3}{5\sqrt{10}}$$

$$= \frac{9}{5\sqrt{10}}$$

$\therefore$ The value of $\cos(A+B) = \frac{9}{5\sqrt{10}}$.

Total: 20 marks

June 2015 – Question 3 – Solution

(a) Evaluate $\lim_{x\to 0}\frac{1-\cos^2(2x)}{x^2}$. [4]

$$\lim_{x\to 0}\frac{1-\cos^2(2x)}{x^2} = \lim_{x\to 0}\frac{\sin^2 2x}{x^2}$$

$$= \lim_{x\to 0}\frac{(\sin 2x)(\sin 2x)}{(x)(x)}$$

$$= \lim_{x\to 0}\left(\frac{\sin 2x}{x}\cdot\frac{\sin 2x}{x}\right)$$

$$= \lim_{x\to 0}\left(\frac{\sin 2x}{x}\right)\times\lim_{x\to 0}\left(\frac{\sin 2x}{x}\right)$$

$$= \lim_{x\to 0}\left(\frac{\sin 2x}{x}\cdot\frac{2}{2}\right)\times\lim_{x\to 0}\left(\frac{\sin 2x}{x}\cdot\frac{2}{2}\right)$$

$$= 2\lim_{x\to 0}\left(\frac{\sin 2x}{2x}\right)\times 2\lim_{x\to 0}\left(\frac{\sin 2x}{2x}\right)$$

$$= 2(1)\times 2(1)$$

$$= 2\times 2$$

$$= 4$$

$$\therefore \lim_{x\to 0}\frac{1-\cos^2(2x)}{x^2} = 4$$

(b) A spherical shaped ice cream is melting in such a way that its volume is decreasing at a rate of 2 cm^3 per minute. Find the rate at which the radius decreases when the diameter is 8 cm. [6]

We are given that $\frac{dV}{dt} = -2cm^3/min.$

$$\text{Radius} = \frac{Diameter}{2}$$

$$= \frac{8}{2}$$

$$= 4\ cm$$

The volume of a sphere is:

$$V = \frac{4}{3}\pi r^3$$

$$\frac{dV}{dr} = \frac{4}{3}(3)\pi r^{3-1}$$

$$= 4\pi r^2$$

When $r = 4$,

$$\frac{dV}{dr} = 4\pi(4)^2$$

$$= 64\pi$$

Using the chain rule,

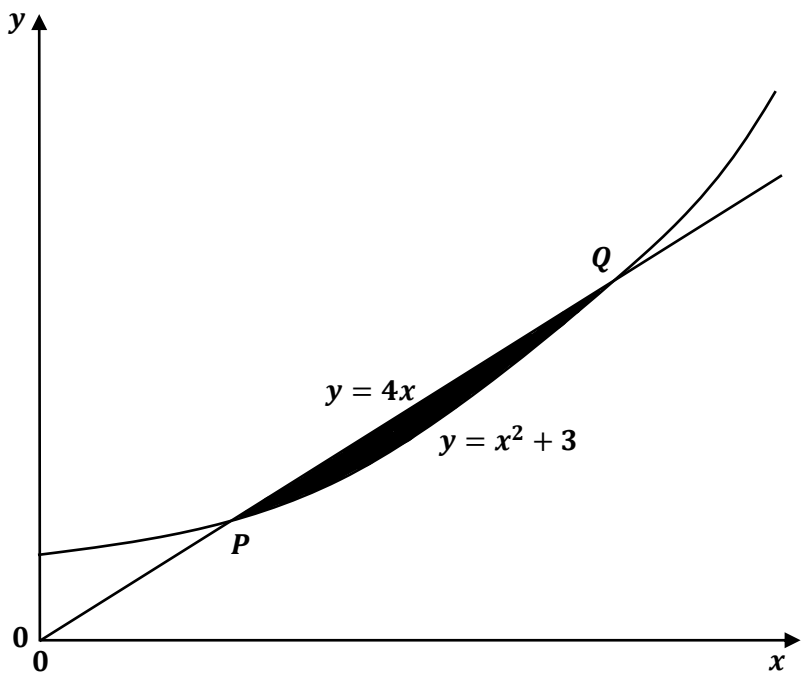
$$\frac{dr}{dt} = \frac{dr}{dV} \times \frac{dV}{dt}$$

$$= \frac{1}{64\pi} \times -2$$

$$= -\frac{1}{32\pi} cm/min$$

$\therefore$ The rate at which the radius decreases when the diameter is 8 cm is $\frac{1}{32\pi} cm/min.$

(c) The diagram below (**not drawn to scale**) is a sketch of the curve $y = x^2 + 3$ and the line $y = 4x$.



(i) Find the coordinates of $\boldsymbol{P}$ and $\boldsymbol{Q}$, the points of intersection of the curve and the line. [5]

$y = x^2 + 3$ → Equation 1

$y = 4x$ → Equation 2

Equating Equation 1 and Equation 2 gives:

$$x^2 + 3 = 4x$$

$$x^2 - 4x + 3 = 0$$

$$(x - 1)(x - 3) = 0$$

Either $x - 1 = 0$ or $x - 3 = 0$

$x = 1$ $x = 3$

When $x = 1$,

$y = 4(1)$

$= 4$

The point is $(1, 4)$.

When $x = 3$,

$y = 4(3)$

$= 12$

The point is $(3, 12)$.

Looking at the diagram, the coordinates are $P(1, 4)$ and $Q(3, 12)$.

(ii) Calculate the volume of the solid generated when the shaded region is rotated completely about the x-axis. [5]

Let V_1 be the volume generated when the line is rotated completely about the x-axis.

Consider,

$y = 4x$

$y^2 = (4x)^2$

$y^2 = 16x^2$

$$V_1 = \pi \int_a^b y^2\, dx$$

$$= \pi \int_1^3 16x^2\, dx$$

$$= 16\pi \int_1^3 x^2\, dx$$

$$= 16\pi \left[\frac{x^3}{3}\right]_1^3$$

$$= \frac{16\pi}{3}[x^3]_1^3$$

$$= \frac{16\pi}{3}[(3)^3 - (1)^3]$$

$$= \frac{16\pi}{3}(27 - 1)$$

$$= \frac{16\pi}{3}(26)$$

$$= \frac{416\pi}{3} \text{ units}^3$$

Let V_2 be the volume generated when the curve is rotated completely about the x-axis.

Consider,

$$y = x^2 + 3$$

$$y^2 = (x^2 + 3)^2$$

$$y^2 = x^4 + 6x^2 + 9$$

Now,

$$V_2 = \pi \int_a^b y^2\, dx$$

$$= \pi \int_1^3 (x^4 + 6x^2 + 9)\, dx$$

$$= \pi \left[\frac{x^5}{5} + 2x^3 + 9x\right]_1^3$$

$$= \pi\left\{\left[\frac{(3)^5}{5} + 2(3)^3 + 9(3)\right] - \left[\frac{(1)^5}{5} + 2(1)^3 + 9(1)\right]\right\}$$

$$= \pi\left[\left(\frac{243}{5} + 54 + 27\right) - \left(\frac{1}{5} + 2 + 9\right)\right]$$

$$= \pi\left(\frac{648}{5} - \frac{56}{5}\right)$$

$$= \frac{592\pi}{5}\text{ units}^3$$

Now,

Volume of the solid generated $= V_1 - V_2$

$$= \frac{416\pi}{3} - \frac{592\pi}{5}$$

$$= \frac{304\pi}{15}\text{ units}^3$$

$\therefore$ The volume of the solid generated when the shaded region is rotated completely about the x-axis is $\frac{304\pi}{15}$ units^3.

Total: 20 marks